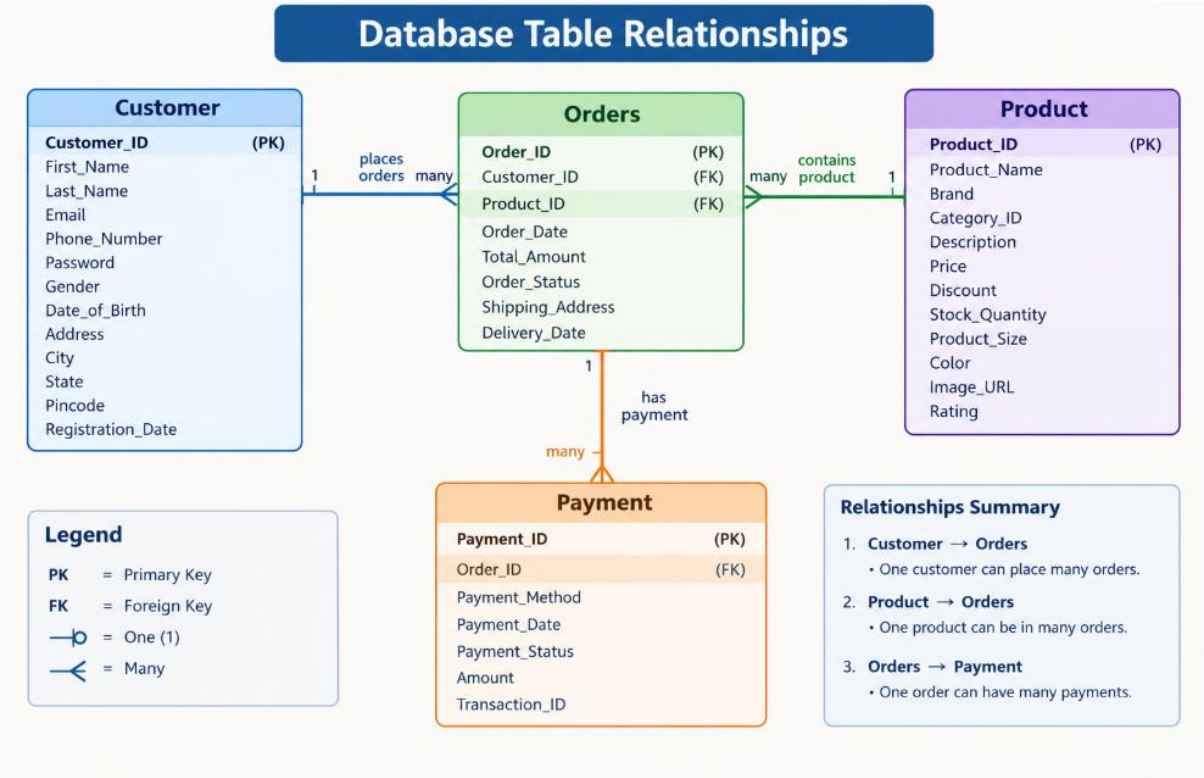
AJIO E-COMMERCE DATABASE MANAGEMENT SYSTEM

DATABASE RELATIONSHIP ANALYSIS USING JOINS

1. DATABASE TABLE RELATIONSHIPS



2. INNER JOIN

```
SQL> SELECT Customer.Customer_ID,
  2         Customer.First_Name,
  3         Orders.Order_ID,
  4         Orders.Total_Amount
  5  FROM Customer
  6  INNER JOIN Orders
  7  ON Customer.Customer_ID = Orders.Customer_ID;

CUSTOMER_ID FIRST_NAME                      ORDER_ID TOTAL_AMOUNT
----------- ------------------------------ ---------- ------------
          1 Pooja                             101         2499
          2 Rahul                             102         1799
          3 Ananya                            103         3299
          4 Arjun                             104         1499
          5 Priya                             105         2899
```

3. LEFT JOIN

```
SQL> SELECT Customer.Customer_ID,
  2         Customer.First_Name,
  3         Orders.Order_ID,
  4         Orders.Total_Amount
  5  FROM Customer
  6  LEFT JOIN Orders
  7  ON Customer.Customer_ID = Orders.Customer_ID;

CUSTOMER_ID FIRST_NAME                    ORDER_ID TOTAL_AMOUNT
----------- ----------------------------- -------- ------------
          1 Pooja                          101         2499
          2 Rahul                          102         1799
          3 Ananya                         103         3299
          4 Arjun                          104         1499
          5 Priya                          105         2899
```

4. RIGHT JOIN

```
SQL> SELECT Customer.First_Name,
  2         Orders.Order_ID,
  3         Orders.Total_Amount,
  4         Orders.Order_Status
  5  FROM Customer
  6  RIGHT JOIN Orders
  7  ON Customer.Customer_ID = Orders.Customer_ID;

FIRST_NAME                     ORDER_ID TOTAL_AMOUNT
------------------------------ -------- ------------
ORDER_STATUS
------------------------------
Pooja                             101         2499
Delivered

Rahul                             102         1799
Shipped

Ananya                            103         3299
Processing


FIRST_NAME                     ORDER_ID TOTAL_AMOUNT
------------------------------ -------- ------------
ORDER_STATUS
------------------------------
Arjun                             104         1499
Delivered

Priya                             105         2899
Shipped
```

5. RETRIEVING COMPLETE ORDER DETAILS

```
SQL> SELECT Customer.First_Name,
  2         Orders.Order_ID,
  3         Orders.Order_Date,
  4         Orders.Total_Amount,
  5         Orders.Order_Status,
  6         Payment.Payment_Method,
  7         Payment.Payment_Status
  8  FROM Customer
  9  JOIN Orders
 10  ON Customer.Customer_ID = Orders.Customer_ID
 11  JOIN Payment
 12  ON Orders.Order_ID = Payment.Order_ID;

FIRST_NAME                     ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS                   PAYMENT_METHOD
------------------------------ ------------------------------
PAYMENT_STATUS
--------------------
Pooja                                 101 10-SEP-26         2499
Delivered                      UPI
Successful

Rahul                                 102 11-SEP-26         1799
Shipped                        Credit Card
Successful

FIRST_NAME                     ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS                   PAYMENT_METHOD
------------------------------ ------------------------------
PAYMENT_STATUS
--------------------

Ananya                                103 12-SEP-26         3299
Processing                     Debit Card
Failed

Arjun                                 104 13-SEP-26         1499
Delivered                      Cash on Delivery

FIRST_NAME                     ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS                   PAYMENT_METHOD
------------------------------ ------------------------------
PAYMENT_STATUS
--------------------
Successful

Priya                                 105 14-SEP-26         2899
Shipped                        UPI
Successful
```

6. CUSTOMER PURCHASE HISTORY

```
SQL> SELECT Customer.Customer_ID,
  2         Customer.First_Name,
  3         Orders.Order_ID,
  4         Orders.Order_Date,
  5         Orders.Total_Amount,
  6         Orders.Order_Status
  7  FROM Customer
  8  JOIN Orders
  9  ON Customer.Customer_ID = Orders.Customer_ID
 10  ORDER BY Customer.Customer_ID;

CUSTOMER_ID FIRST_NAME                     ORDER_ID ORDER_DAT TOTAL_AMOUNT
----------- ------------------------------ ---------- --------- ------------
ORDER_STATUS
------------------------------
          1 Pooja                               101 10-SEP-26         2499
Delivered

          2 Rahul                               102 11-SEP-26         1799
Shipped

          3 Ananya                              103 12-SEP-26         3299
Processing


CUSTOMER_ID FIRST_NAME                     ORDER_ID ORDER_DAT TOTAL_AMOUNT
----------- ------------------------------ ---------- --------- ------------
ORDER_STATUS
------------------------------
          4 Arjun                               104 13-SEP-26         1499
Delivered

          5 Priya                               105 14-SEP-26         2899
Shipped
```

7. MULTI-TABLE REPORT

```
SQL> SELECT Customer.First_Name,
  2         Orders.Order_ID,
  3         Orders.Total_Amount,
  4         Orders.Order_Status,
  5         Payment.Payment_Method,
  6         Payment.Payment_Status,
  7         Payment.Transaction_ID
  8  FROM Customer
  9  JOIN Orders
 10  ON Customer.Customer_ID = Orders.Customer_ID
 11  JOIN Payment
 12  ON Orders.Order_ID = Payment.Order_ID;
```

```
FIRST_NAME                       ORDER_ID TOTAL_AMOUNT
------------------------------ ---------- ------------
ORDER_STATUS                   PAYMENT_METHOD
------------------------------ ------------------------------
PAYMENT_STATUS       TRANSACTION_ID
-------------------- --------------------------------------------------
Pooja                                 101         2499
Delivered                      UPI
Successful           TXN1001

Rahul                                 102         1799
Shipped                        Credit Card
Successful           TXN1002

FIRST_NAME                       ORDER_ID TOTAL_AMOUNT
------------------------------ ---------- ------------
ORDER_STATUS                   PAYMENT_METHOD
------------------------------ ------------------------------
PAYMENT_STATUS       TRANSACTION_ID
-------------------- --------------------------------------------------

Ananya                                103         3299
Processing                     Debit Card
Failed               TXN1003

Arjun                                 104         1499
Delivered                      Cash on Delivery

FIRST_NAME                       ORDER_ID TOTAL_AMOUNT
------------------------------ ---------- ------------
ORDER_STATUS                   PAYMENT_METHOD
------------------------------ ------------------------------
PAYMENT_STATUS       TRANSACTION_ID
-------------------- --------------------------------------------------
Successful           TXN1004

Priya                                 105         2899
Shipped                        UPI
Successful           TXN1005
```